DOCUMENT CONTROL & METADATA			
Project Name:	Thermal Analysis for Air-Dryer Chamber	**Lead Analyst:**	AVX Engineering Team
Simulation Type:	CFD and Thermal Analysis	**Software Tool:**	SolidWorks 2026
Revision:	v1.0	**Date**	Sept. 14, 2026

1. Introduction

This technical assessment was conducted to evaluate the thermal and fluid behavior of the proposed system through numerical simulation analysis. The assessment utilized advanced simulation software to analyze heat transfer characteristics, temperature distribution, and fluid flow behavior under defined operating conditions.

The primary objective of this study is to determine whether the system can achieve and maintain the required operating temperature based on the baseline Bill of Materials (BOM) specifications, particularly for the inductors and other critical components involved. The analysis aims to verify that the selected materials and component configurations are capable of withstanding the expected thermal conditions while ensuring reliable operation and performance.

The simulation considers key thermal and fluid parameters, including heat generation, heat dissipation, temperature gradients, and fluid interaction within the system. By evaluating these factors, the assessment provides insights into potential thermal limitations, areas of improvement, and the overall feasibility of the proposed design.

The results obtained from this analysis will serve as a technical reference for validating the current design approach, optimizing system performance, and ensuring that all components operate within their specified temperature limits. This assessment supports informed engineering decisions by identifying possible risks related to overheating, thermal stress, and material degradation before physical prototyping or implementation.

2. Requirements

Preliminary	
Load	15 kg dying capacity
No. of Trays	3
Operating Temp.	40-55 C deg
Airflow	30-60 CFM
Heat Flux	1800 W

3. Simulation Setup

SolidWorks CFD	
Analysis Type	Internal
Unit System	SI (m-kg-s)
Fluid	Air
Total Cell Counts Mesh	233,564
Flow Type	Laminar and Turbulent
Gravity	On
No. of Iterations	450
Calculation Time	1.3 Hours
Boundary Conditions	

SolidWorks CFD	
Inlet Volume Flow	0.00236 m³/s or 5 CFM (adjusted from iterations)
Environment Pressure	101325 Pascal
Heat Flux	500 W (adjusted from iterations)

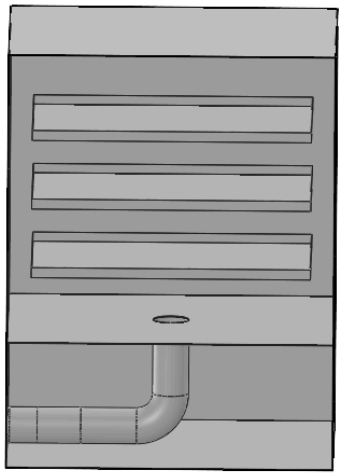


Figure 1. SolidWorks 3D Model

Presents the 3D solid geometry used to set up the CFD fluid domain. It defines all internal pipe surfaces, elbows, and diffuser outlets for computational meshing.

4. Simulation Report

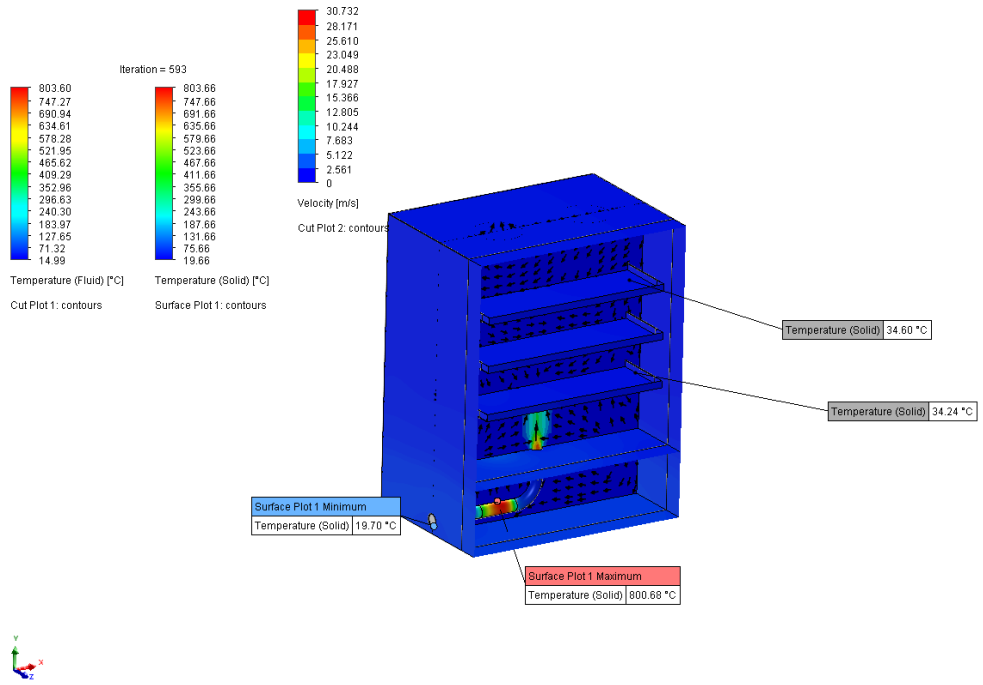


Figure 2. Iteration 60 CFM - 750 Watts

The system can operate more efficiently by reducing the airflow rate (CFM), which will minimize blower power consumption. If a higher chamber temperature is required, the heat generator power can be increased instead of increasing the airflow rate.

Increasing the CFM while reducing the heat generator power may result in a lower overall chamber temperature. This is because a higher airflow rate increases the fluid velocity inside the tube, causing the generated heat to be removed more rapidly through convection. As a result, the increased cooling effect from the airflow reduces the amount of heat retained inside the chamber.

Therefore, optimizing the balance between airflow rate and heat generator power is essential to achieve the desired chamber temperature while maintaining energy efficiency and minimizing blower power consumption.

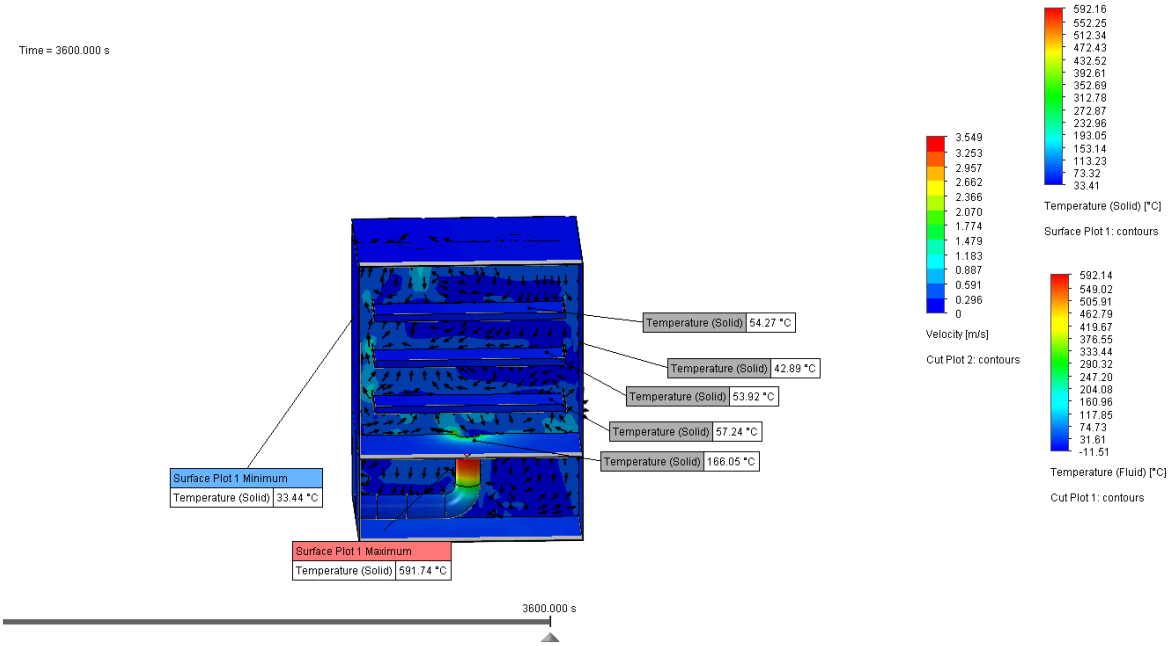


Figure 3. Iteration 5 CFM - 500 Watts

By minimizing the CFM, the simulation results demonstrate that the chamber is capable of reaching and maintaining the required operating temperature while sustaining heat retention over an extended period. The reduced airflow rate helps minimize excessive heat removal, allowing the generated thermal energy to remain within the chamber and maintain a stable temperature condition.

At the same time, the optimized CFM and heat flux power values ensure that the system operates within the maximum allowable temperature limits of the materials involved. This prevents potential thermal degradation or damage while maintaining sufficient heating performance.

Based on the analysis results, the selected CFM and heat generator heat flux power represent the most optimized operating parameters, providing an effective balance between thermal performance, energy consumption, and material safety. These values are therefore recommended as the baseline operating conditions for achieving reliable and efficient chamber temperature control.

Tabular Results @1hr (3600 s)	
Chamber Temp.	50 deg C
Tube Temp.	591.74 deg C
Tray 1 Temp. (Bottom)	57.24 deg C
Tray 2 Temp. (MIddle)	53.92 deg C
Tray 3 Temp. (Top)	54.27 deg C

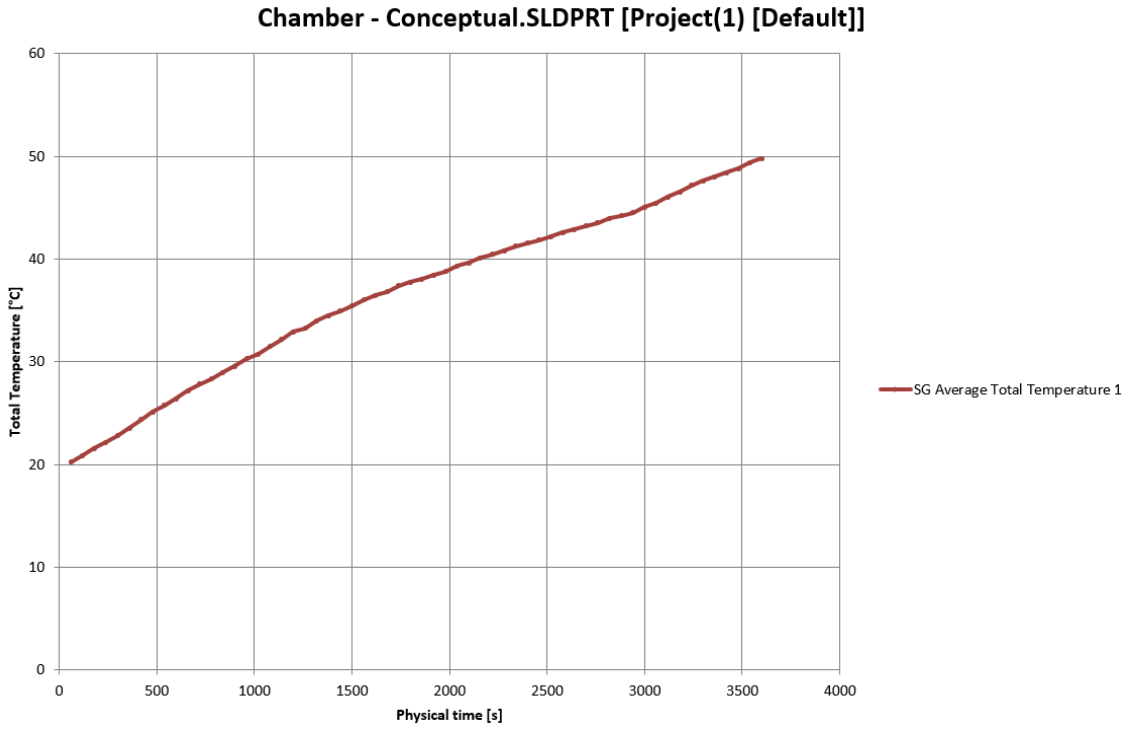


Figure 4. Graph Temp. vs Time

The graph illustrates the temperature variation over time obtained from the transient simulation analysis conducted for a duration of one (1) hour. It presents the temperature response of the system at each time interval, allowing the evaluation of the heating behavior, temperature rise rate, and stabilization period within the chamber.

Based on the simulation results, the temperature trend can be used to verify whether the selected CFM and heat flux power settings are sufficient to sustain the required operating

temperature while preventing the system from exceeding the maximum allowable temperature limits of the materials. This confirms the reliability of the optimized parameters for continuous operation and thermal management.

Video Source: https://youtu.be/4NUgGcm0AFE

Chamber Materials	
Frame and Trays	Stainless Steel 304
Inductor Pipe	AISI 1018 Low Carbon Steel or Similar; 3” Diameter
Insulator	Polyurethane

5. Detailed Drawing

Dimensions	
Frame SIze	550 x 700 x 900 mm
Tray Size	600 x 400 mm
Chamber Size	704 x 630 x 548 mm
Inlet Diameter	3 inches pipe

6. Analysis

The thermal and fluid simulation was performed through multiple iterations to optimize the airflow rate, heat flux power, and tube geometry. The initial evaluation using **30 CFM** showed excessive airflow velocity, which may cause the rice samples on the trays to be displaced. Increasing the airflow to **60 CFM** further increased this risk while also reducing thermal performance.

The higher airflow conditions resulted in increased cooling inside the tube due to the higher volume of ambient-temperature air entering the system. This caused greater convective heat loss, preventing the chamber from reaching the required operating temperature even with approximately **2.2 kW heat flux power**.

Additional iterations were conducted by varying the tube diameter. Smaller tube diameters increased airflow velocity and dynamic pressure, creating a higher risk of disturbing the tray contents. The **3-inch tube diameter** was selected as it provided a better balance between airflow distribution and velocity control. The optimized simulation result was achieved at **5 CFM with 500 Watts Heat Inductor**, where the chamber was able to reach and maintain the required operating temperature with improved thermal efficiency.

7. Recommendation

Based on the simulation results, the recommended operating condition is a **3-inch tube diameter with 5 CFM airflow and optimized heat flux power of 500 Watts**. This configuration provides the best balance between heating performance, airflow stability, and energy efficiency.

The reduced CFM minimizes blower power consumption while preventing excessive cooling effects and maintaining the required chamber temperature. This optimized configuration is recommended as the baseline parameter for prototype testing and further validation.

In the next page, the suggested updated materials are as follows and system block diagram.

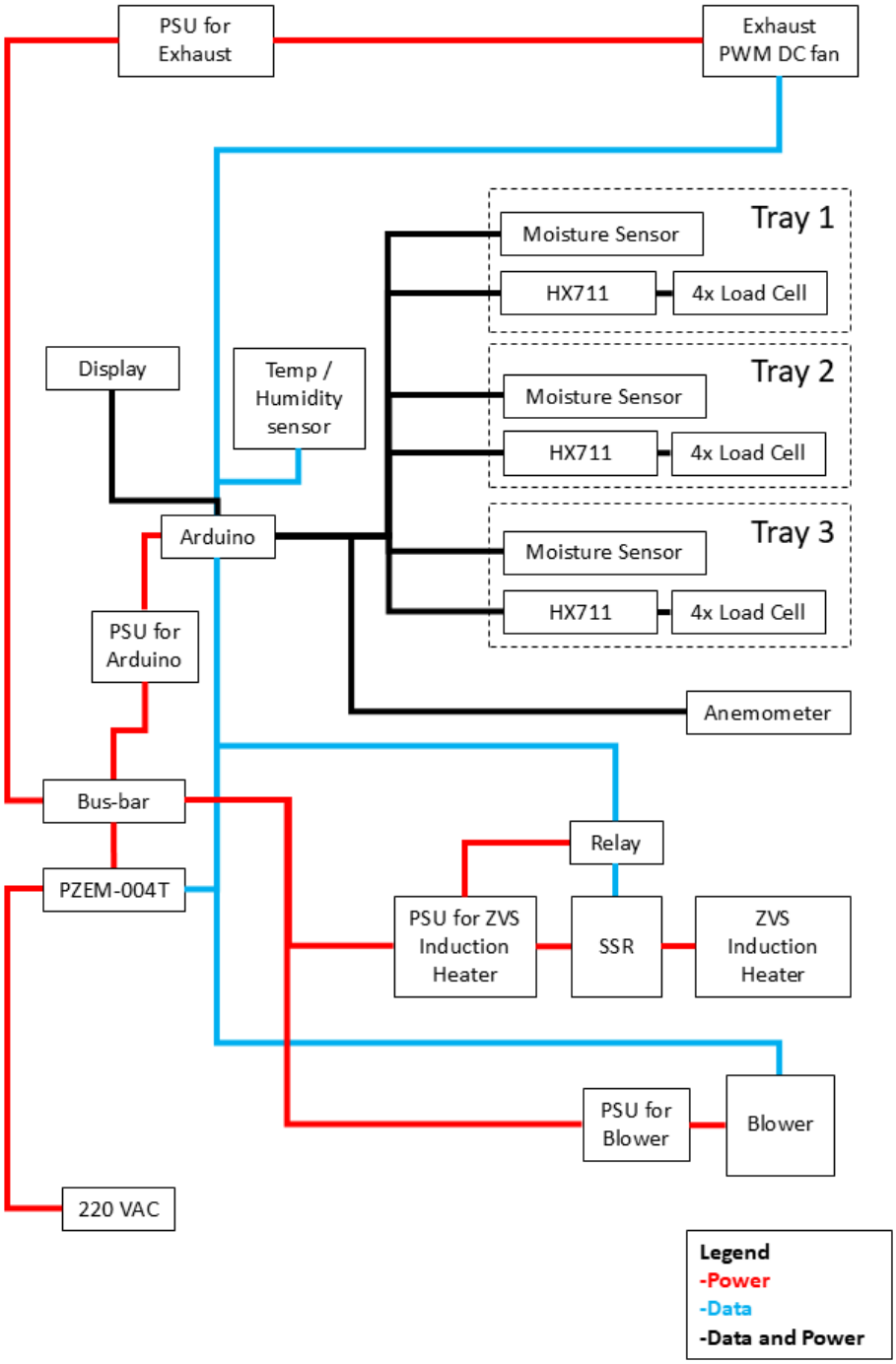
Item No.	Status	Component	Description	Specification	Remarks	Alternative
1	Initial	DS18B20 Temperature Sensor	High-precision waterproof digital probe used to monitor the target 40C to 60C internal chamber environment.	Temperature sensor supply voltage: 3.0V ~ 5.5V Temperature range: -55 ~ +125 °	OK, alternative could be SHT40 for temp and humidity	SHT40 probe supply voltage: 1.08 - 3.6 V interface: I2C temperature accuracy: ±0.2°C (0-65°C) temperature range: -40 - 125°C humidity accuracy: ±1.8% RH (30-70RH at 25°C) relative humidity range: 0 - 100 %RH
2	Initial	DHT22 Sensor	High-accuracy digital hygrometer to monitor internal relative air vapor saturation.	Supply voltage: 3 to 5V 0-100% humidity readings with 2-5% accuracy -40 to 80°C temperature readings ±0.5°C accuracy	OK, alternative could be SHT40 for temp and humidity	
3	Additional	Level Shifter			Level Shifter for SHT40 to convert 3.3v sensor to 5v arduino.	
4	Additional	AMS1117-3.3 Power Supply Module			AMS1117-3.3 Power Supply Module for 3.3v sensors and level shifter	
5	Initial	RS485 Modbus Moisture Sensor Module – Moisture Content (MC) Sensor	Moisture tracking module placed in the grain bed to measure real-time moisture content reduction.	Supply voltage: 5-30V Communication: RS485 Probe Type: 3Pin	1 sensor per tray, recommended to maintain probes surrounded by grains to avoid measuring ambient, to test and calibrate in actual implementation, alternative could be	Capacitive moisture sensor

					a capacitive moisture sensor	
6	Additional	RS485 converter			Include TTL to RS485 converter	
7	Initial	PZEM-004T	AC power monitor used to track cumulative electricity consumption for Specific Energy Consumption (SEC) calculations.	Operating Voltage: 80-260VAC Test Voltage: 80-260VAC Rated Current: 100A Rated power: 100A / 22000W	OK	
8	Initial	High-Frequency Electromagnetic Induction Heater	The primary heating driver used to induce eddy currents into the medium	ZVS High Frequency Induction Heating Module Power: 1800W Voltage: 12-48V	OK but note that ZVS modules has no modulation control to adjust temperature, it is also not recommended to use PWM on power as the high current draw can easily damage the power supply. Recommendation to use 500W as baseline and additional smaller modules to increase heating capacity as needed. Use SSR in between the ZVS module and power supply to turn on and off. Recommended to select either 18v or 24v model.	
9	Additional				Solid State Relay (SSR) for ZVS	

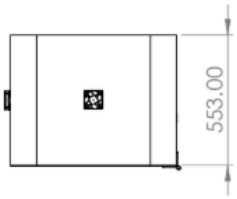
					Induction heater with 2.0 SF from ZVS max current draw	
10	Additional				DC PSU for ZVS Induction heater	
11	Additional				DC PSU for SSR activation, assuming it is higher than 5v, could also use the ZVS PSU for activation if compatible.	
12	Initial	Anemometer	A high-precision sensor instrument deployed at the drying chamber to quantify the volumetric and linear velocity of air moving through the	DC Power Supply: 5-30V DC Measuring Range: 0-70m/s	OK but note that operating temperature of the heating chamber may damage plastic components	
13	Initial	Motor Blower (Electric Air Blower)	Used to establish continuous forced convection within the drying chamber.	Voltage: 220V-240V-50/60Hz Input power:400W No-load speed: 14000rpm Max blowing rate: 3.0m3/min	No modulation control built-in the AC blower. Recommendation to use DC fan and a driver, another recommendation is to use PWM DC fan for simplified integration to arduino.	PWM DC airflow fan
14	Initial	Load Cell Amplifier HX711	The HX711 is a precision 24-bit analog-to-digital converter (ADC) and amplifier designed for weigh scales	Operation Voltage: 2.7V--5V Operation Current: < 1.5mA Selectable 10SPS or	OK, 1 amplifier per tray	

			and industrial control applications	80SPS output data rate Simultaneous 50 and 60Hz supply rejection		
15	Initial	Straight Bar Load Cell Weight Sensor	A straight bar load cell is a strain-gauge-based sensor that translates applied force or weight into electrical signals.	Model: YZC-131 Material: Aluminum Dimension: 75x12.7x12.7mm Range: 1 - 10kg Rated Output: 1.0± 0.1mV/V Operating Temperature Range: -20 - 65°C	OK, 4 load cell per tray	
16	Initial	Arduino Mega 2560 R3	The main controller that handles all sensor inputs, executes the automated logic loop, and control the outputs.	54 digital input/output pins 256 KB of Flash memory for code storage 8 KB of SRAM for data	OK	
17	Additional	Arduino 5v PSU			PSU for Arduino	
18	Initial	20x4 LCD	Used to display real-time values of temperature, humidity, moisture content, and energy consumption.	HD44780 parallel interface chipset Green LED Backlight	OK	
19	Initial	Relay	Interfaced with the Arduino Mega to switch power to the induction heater and	Relay Voltage: 5V	OK, relay1 for SSR, Optional / depends on selected set-up: relay2 for AC Blower fan, relay3	

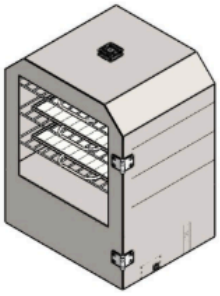
			motor blower safely based on the temperature parameters.		for AC Exhaust Fan	
20	Initial	Exhaust Fan	Output vents of the drying chamber.	Voltage/frequency: 200V-240/50-60 HZ Power: 20W Air volume: 150m³/h Speed: 2800r/min	No modulation control built-in the AC fan. Recommendation to use DC fan and a driver, another recommendation is to use PWM DC fan for simplified integration to arduino.	PWM DC airflow fan
21	Additional				PSU for PWM DC fan	
22	Additional				Miscellaneous parts for bus-bar, connectors, protoboard, case, wires, and other consumables.	


PSU for Exhaust
Exhaust PWM DC fan
Tray 1
Moisture Sensor
HX711
4x Load Cell
Tray 2
Moisture Sensor
HX711
4x Load Cell
Tray 3
Moisture Sensor
HX711
4x Load Cell
Display
Temp / Humidity sensor
Arduino
PSU for Arduino
Anemometer
Bus-bar
Relay
PZEM-004T
PSU for ZVS Induction Heater
SSR
ZVS Induction Heater
PSU for Blower
Blower
220 VAC
Legend
-Power
-Data
-Data and Power

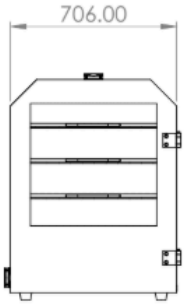
8. Appendix



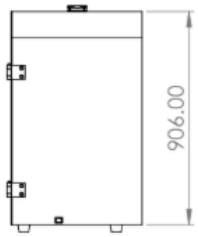
TOP VIEW



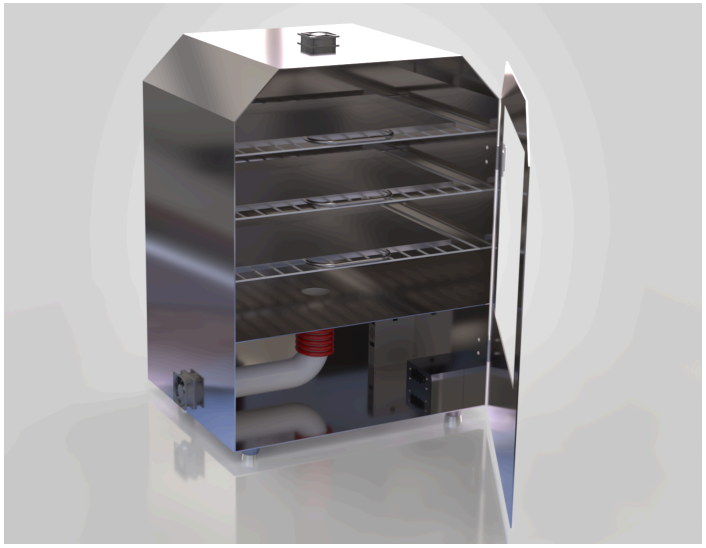
ISOMETRIC VIEW



FRONT VIEW



SIDE VIEW



NOTHING FOLLOWS